

Data Cleaning Part 2

Data Wrangling in R

Data Cleaning Part 2

Example of Cleaning: more complicated

For example, let's say we have a variable about treatment or control conditions coded as treatment, T, treat, Treat, C, Cont, cont, cOnt, Control, and control. Using Excel to find all of these would be a matter of filtering and changing all by hand or using if statements.

Sometimes though, it's not so simple. That's where functions that find patterns come to be very useful.

Take a look at the data

```
count(data_gen, status)
```

```
# A tibble: 11 × 2
  status      n
  <chr>    <int>
1 C         81
2 Cont      90
3 Control   91
4 T         91
5 Traet    105
6 Treat    100
7 cOnt      79
8 cont      83
9 control   98
10 treat     86
11 treatment 96
```

Example of Cleaning: more complicated

In R, you could use `case_when()`:

```
#case_when way:
data_gen <- data_gen %>% mutate(status =
  case_when(status
    %in% c("C", "cont", "cOnt", "Cont", "control", "Control")
          ~ "Control",
    .default = status))
count(data_gen, status)
```

```
# A tibble: 6 × 2
  status      n
  <chr>    <int>
1 Control    522
2 T          91
3 Traet     105
4 Treat     100
5 treat      86
6 treatment  96
```

Oh dear! This only fixes some values! It is difficult to notice values like "Traet".

String functions

The **stringr** package

Like `dplyr`, the `stringr` package:

- Makes some things more intuitive
- Is different than base R
- Is used on forums for answers
- Has a standard format for most functions: `str_`
 - the first argument is a string like first argument is a `data.frame` in `dplyr`

Useful String Functions

Useful String functions from base R and `stringr`

- `toupper()`, `tolower()` - uppercase or lowercase your data
- `str_sentence()` - uppercase just the first character (in the `stringr` package)
- `paste()` - paste strings together with a space
- `paste0` - paste strings together with no space as default
- `str_trim()` (in the `stringr` package) or `trimws` in base
 - will trim whitespace
- `nchar` - get the number of characters in a string

recoding with `str_to_sentence()`

```
#case_when way:  
data_gen <- data_gen %>%  
  mutate(status = str_to_sentence(status))  
count(data_gen, status)
```

```
# A tibble: 5 × 2  
  status      n  
  <chr>    <int>  
1 Control    522  
2 T          91  
3 Traet     105  
4 Treat     186  
5 Treatment   96
```

recoding with `str_to_sentence()`

```
#case_when way:
data_gen <-data_gen %>%
  mutate(status = str_to_sentence(status)) %>%
  mutate(status =
    case_when(status %in%
      c("Treatment", "T", "Treat", "Traet", "Treat")
        ~ "Treatment",
      .default = status))
count(data_gen, status)
```

```
# A tibble: 2 × 2
  status      n
  <chr>    <int>
1 Control    522
2 Treatment  478
```

OK, now we are getting somewhere!

Reading in again

Now we have a chance to keep but clean these values!

```
ufo <-read_csv(  
  "https://sisbid.github.io/Data-Wrangling/data/ufo/ufo_data_complete.csv",  
  col_types = cols(`duration (seconds)` = "c"))
```

Warning: One or more parsing issues, call `problems()` on your data frame for
dat <- vroom(...)
problems(dat)

Clean names with the `clean_names()` function from the **janitor** package

```
colnames(ufo)
```

```
[1] "datetime"      "city"           "state"
[4] "country"       "shape"          "duration (seconds)"
[7] "duration (hours/min)" "comments"       "date posted"
[10] "latitude"      "longitude"
```

```
ufo_clean <- clean_names(ufo)
colnames(ufo_clean)
```

```
[1] "datetime"      "city"           "state"
[4] "country"       "shape"          "duration_seconds"
[7] "duration_hours_min" "comments"       "date_posted"
[10] "latitude"      "longitude"
```

str_detect and filter

Now let's fix our ufo data and remove those pesky backticks in the `duration_seconds` variable. First let's find them with `str_detect`.

```
ufo_clean %>%  
  filter(str_detect(  
    string = duration_seconds,  
    pattern = "`"))
```

```
# A tibble: 3 × 11  
  datetime          city state country shape duration_seconds duration_hours_r  
  <chr>             <chr> <chr> <chr>   <chr> <chr>              <chr>  
1 2/2/2000 19:33 bouse az us <NA> 2` each a few second  
2 4/10/2005 22:52 santa... ca us <NA> 8` eight seconds  
3 7/21/2006 13:00 ibagu... <NA> <NA> circ... 0.5` 1/2 segundo  
# i 4 more variables: comments <chr>, date_posted <chr>, latitude <chr>,  
# longitude <chr>
```

str_remove

```
ufo_clean <- ufo_clean %>%  
  mutate(duration_seconds =  
    str_remove(string = duration_seconds,  
               pattern = "`"))
```

Lets also mutate to be as.numeric again

```
ufo_clean <- ufo_clean %>%
  mutate(duration_seconds = as.numeric(duration_seconds))
```

```
glimpse(ufo_clean)
```

Rows: 88,875

Columns: 11

```
$ datetime      <chr> "10/10/1949 20:30", "10/10/1949 21:00", "10/10/1955 21:00"
```

```
$ city <chr> "san marcos", "lackland afb", "chester" (uk/england
```

```
$ state      <chr> "tx", "tx", NA, "tx", "hi", "tn", NA, "ct", "al", "
```

```
$ country      <chr> "us", NA, "gb", "us", "us", "us", "gb", "us", "us"
```

```
$ shape <chr> "cylinder", "light", "circle", "circle", "light", "
```

```
$ duration_seconds <dbl> 2700, 7200, 20, 20, 900, 300, 180, 1200, 180, 120,
```

```
$ duration_hours_min <chr> "45 minutes", "1-2 hrs", "20 seconds", "1/2 hour",
```

```
$ comments      <chr> "This event took place in early fall around 1949-50"
$ date          <chr> "1/27/2000"      "12/12/2000"      "1/21/2000"      "1/17/2000"
```

```
$ date_posted <chr> "4/27/2004", "12/16/2005", "1/21/2008", "1/17/2004"
```

```
$ latitude      <chr> "29.8830556", "29.38421", "53.2", "28.9783333", "21.11111"
```

```
$ longitude      <chr> "-97.9411111", "-98.581082", "-2.916667", "-96.6458"
```

Substringing

stringr

- `str_sub(x, start, end)` - substrings from position start to position end

Substringing

Examples:

```
str_sub("I like friesian horses", 8,12)
```

```
[1] "fries"
```

```
#123456789101112
```

```
#I like fries
```

```
str_sub(c("Site A", "Site B", "Site C"), 6,6)
```

```
[1] "A" "B" "C"
```

Splitting/Find/Replace and Regular Expressions

- R can do much more than find exact matches for a whole string
- Like Perl and other languages, it can use regular expressions.
- What are regular expressions?
 - Ways to search for specific strings
 - Can be very complicated or simple
 - Highly Useful - think “Find” on steroids

A bit on Regular Expressions

- <http://www.regular-expressions.info/reference.html>
- They can use to match a large number of strings in one statement
- `.` matches any single character
- `*` means repeat as many (even if 0) more times the last character
- `?` makes a pattern optional (i.e. it matches 0 or 1 times)
- `^` matches start of vector `^a` - starts with "a"
- `$` matches end of vector `b$` - ends with "b"

'Find' functions: **stringr**

`str_detect`, `str_subset`, `str_replace`, and `str_replace_all` search for matches to argument pattern within each element of a character vector: they differ in the format of and amount of detail in the results.

- `str_detect` - returns TRUE if pattern is found
- `str_subset` - returns only the strings where the pattern were detected
- `str_extract` - returns only the pattern that was detected
- `str_replace` - replaces pattern with replacement the first time
- `str_replace_all` - replaces pattern with replacement as many times matched

'Find' functions: Finding Indices

These are the indices where the pattern match occurs:

```
ufo_clean %>%  
  filter(str_detect(comments, "two aliens")) %>%  
  head()
```

```
# A tibble: 2 × 11  
  datetime          city state country shape duration_seconds duration_hours_r  
  <chr>            <chr> <chr> <chr>   <chr>          <dbl> <chr>  
1 10/14/2006 02:00 yuma  va    us    form...        300 5 minutes  
2 7/1/2007 23:00  nort... ct    <NA>   unkn...         60 1 minute  
# i 4 more variables: comments <chr>, date_posted <chr>, latitude <chr>,  
#   longitude <chr>
```

To Take a look at comments... need to select it first

```
ufo_clean %>%  
  filter(str_detect(comments, "two aliens")) %>%  
  select(comments)
```

```
# A tibble: 2 × 1
```

```
  comments
```

```
<chr>
```

```
1 ((HOAX??)) two aliens appeared from a bright light to peacefully investigate
```

```
2 Witnessed two aliens walking along baseball field fence.
```

'Find' functions: `str_subset()` is easier

`str_subset()` gives the values that match the pattern: (or if we used `negate = TRUE` it would find the opposite)

```
ufo_clean %>% pull(comments) %>%  
  str_subset( "two aliens")
```

```
[1] "((HOAX??)) two aliens appeared from a bright light to peacefully investigate the surroundings in the woods"  
[2] "Witnessed two aliens walking along baseball field fence."
```

Showing difference in `str_extract`

`str_extract` extracts just the matched string

```
ufo_clean %>%  
  mutate(aliens = str_extract(comments, "two aliens")) %>%  
  count(aliens)
```

```
# A tibble: 2 × 2  
  aliens      n  
  <chr>    <int>  
1 two aliens      2  
2 <NA>      88873
```

- Look for any comment that starts with “aliens”

```
ufo_clean %>% pull(comments) %>% str_subset( "^aliens")
```

```
[1] "aliens speak german???" "aliens exist"          "aliens in srilanka"
```


Using Regular Expressions

That contains space then ship maybe with stuff in between

```
ufo_clean %>% pull(comments) %>%  
  str_subset("space.?ship") %>% head(4) # gets "spaceship" or "space ship" or...
```

```
[1] "I saw the cylinder shaped looked like a spaceship hovering above the east side of the Air Force base. Saw it for  
[2] "description of a spaceship spotted over Birmingham Alabama in 1967."  
[3] "A space ship was descending to the ground"  
[4] "On Monday october 3rd 2005 I spotted two spaceships in the sky. The first spotted ship was what seemed to be a
```

```
ufo_clean %>% pull(comments) %>%  
  str_subset("space.ship") %>% head(4) # no "spaceship" must have character in between
```

```
[1] "A space ship was descending to the ground"  
[2] "I saw a Silver space ship rising into the early morning sky over Houston Texas."  
[3] "Saw a space ship hanging over the southern (Manzano) portion of the Sandia Mountains on evening. It was bright  
[4] "saw space ship for 5 min Got scared crapless"
```

time information

```
pull(ufo_clean, duration_hours_min) %>% head(n = 20)
```

[1]	"45 minutes"	"1-2 hrs"	"20 seconds"	"1/2 hour"
[5]	"15 minutes"	"5 minutes"	"about 3 mins"	"20 minutes"
[9]	"3 minutes"	"several minutes"	"5 min."	"3 minutes"
[13]	"30 min."	"3 minutes"	"30 seconds"	"20minutes"
[17]	"2 minutes"	"20-30 min"	"20 sec."	"2 min"

str_replace()

Let's say we wanted to make the time information more consistent. Using `case_when()` could be very tedious and error-prone!

We can use `str_replace()` to do so.

```
ufo_clean %>% mutate(duration_hours_min =  
  str_replace(string = duration_hours_min,  
    pattern = "minutes",  
    replacement = "mins")) %>%  
  pull(duration_hours_min) %>%  
  head(8)
```

```
[1] "45 mins"      "1-2 hrs"      "20 seconds"   "1/2 hour"     "15 mins"  
[6] "5 mins"       "about 3 mins" "20 mins"
```

Separating columns

Better yet, you might notice that this data isn't tidy- there are more than two entries for each value - amount of time and unit. We could separate this using `separate()` from the `tidyr` package.

```
ufo_clean %>% separate(duration_hours_min,  
                        into = c("duration_amount", "duration_unit"),  
                        sep = " ") %>%  
  select(duration_amount, duration_unit) %>% head()
```

```
# A tibble: 6 × 2  
  duration_amount duration_unit  
    <chr>         <chr>  
1 45            minutes  
2 1-2           hrs  
3 20            seconds  
4 1/2           hour  
5 15            minutes  
6 5             minutes
```

As you can see there is still plenty of cleaning to do!

more seperating

```
ufo_clean <- ufo_clean %>% separate(datetime,  
                                     into = c("date", "time"),  
                                     sep = " ")  
ufo_clean %>% select(date, time) %>% head()
```

```
# A tibble: 6 × 2  
  date      time  
  <chr>    <chr>  
1 10/10/1949 20:30  
2 10/10/1949 21:00  
3 10/10/1955 17:00  
4 10/10/1956 21:00  
5 10/10/1960 20:00  
6 10/10/1961 19:00
```

Dates and times

The [lubridate](<https://lubridate.tidyverse.org/>) package is amazing for dates. Most important functions are those that look like ymd or mdy etc. They specify how a date should be interpreted.

```
library(lubridate)#need to load this one!
```

```
ufo_clean <- ufo_clean %>% mutate(date = mdy(date))  
head(ufo_clean)
```

```
# A tibble: 6 × 12  
  date      time city state country shape duration_seconds duration_hours_min  
  <date>    <chr> <chr> <chr> <chr>  <chr>          <dbl> <chr>  
1 1949-10-10 20:30 san ... tx    us    cyli...      2700 45 minutes  
2 1949-10-10 21:00 lack... tx    <NA>    light      7200 1-2 hrs  
3 1955-10-10 17:00 ches... <NA>  gb     circ...       20 20 seconds  
4 1956-10-10 21:00 edna  tx    us     circ...       20 1/2 hour  
5 1960-10-10 20:00 kane... hi    us     light      900 15 minutes  
6 1961-10-10 19:00 bris... tn    us     sphe...      300 5 minutes  
# i 4 more variables: comments <chr>, date_posted <chr>, latitude <chr>,  
#   longitude <chr>
```

str_*functions

```
str_detect(string = c("abcdd", "two"), pattern = "dd")
```

```
[1] TRUE FALSE
```

```
str_subset(string = c("abcdd", "two"), pattern = "dd")
```

```
[1] "abcdd"
```

```
str_extract(string = c("abcdd", "two"), pattern = "dd")
```

```
[1] "dd" NA
```

```
str_sub(string = c("abcdd", "two"), start = 1, end = 3)
```

```
[1] "abc" "two"
```

Summary

- `stringr` package has lots of helpful functions that work on vectors or variables in a data frame
- `str_detect` helps find patterns
- `str_detect` and `filter` can help you filter data based on patterns within value
- `str_extract` helps extract a pattern
- `str_sub` extracts pieces of strings based on the position of the the characters
- `str_subset` gives the values that match a pattern
- `separate` can separate columns into two
- `^` indicates the start of a string
- `$` indicates the end of a string
- the `lubridate` package is useful for dates and times

Lab

<https://sisbid.github.io/Data-Wrangling/labs/data-cleaning-lab-part2.Rmd>